

```
emertxe@emertxe-VirtualBox: ~/Downloads/RTOS_Codes
emertxe@emertxe-VirtualBox:~/Downloads/RTOS_Codes$ ls
binary_semaphore.c  Commands.txt  critical_section_fixed.c  mutex_demo.c  process_demo.c  race_condition.c  thread_demo.c
emertxe@emertxe-VirtualBox:~/Downloads/RTOS_Codes$ vim process_demo.c
emertxe@emertxe-VirtualBox:~/Downloads/RTOS_Codes$ gcc process_demo.c -o process_demo
emertxe@emertxe-VirtualBox:~/Downloads/RTOS_Codes$ ./process_demo
Before fork()
Parent Process
Parent PID: 5212
Child PID from parent: 5213
Process execution complete
Child Process
Child PID: 5213
Process execution complete
emertxe@emertxe-VirtualBox:~/Downloads/RTOS_Codes$ gcc thread_demo.c -o thread_demo -lpthread
emertxe@emertxe-VirtualBox:~/Downloads/RTOS_Codes$ ./thread_demo
Main Program Started
Main Running: 1
Thread Running: 1
Main Running: 2
Thread Running: 2
Main Running: 3
Thread Running: 3
Main Running: 4
Thread Running: 4
Thread Running: 5
Main Running: 5
Main Program Ended
emertxe@emertxe-VirtualBox:~/Downloads/RTOS_Codes$
```

```
emertxe@emertxe-VirtualBox: ~/Downloads/RTOS_Codes
emertxe@emertxe-VirtualBox:~/Downloads/RTOS_Codes$ gcc race_condition.c -o race_condition -lpthread
emertxe@emertxe-VirtualBox:~/Downloads/RTOS_Codes$ ./race_condition
Final Counter Value: 129328
emertxe@emertxe-VirtualBox:~/Downloads/RTOS_Codes$ ./race_condition
Final Counter Value: 103050
emertxe@emertxe-VirtualBox:~/Downloads/RTOS_Codes$ gcc critical_section_fixed.c -o critical_section_fixed -lpthread
emertxe@emertxe-VirtualBox:~/Downloads/RTOS_Codes$ ./critical_section_fixed
Final Counter Value: 200000
emertxe@emertxe-VirtualBox:~/Downloads/RTOS_Codes$ gcc mutex_demo.c -o mutex_demo -lpthread
emertxe@emertxe-VirtualBox:~/Downloads/RTOS_Codes$ ./mutex_demo
Thread 2: Counter = 1
Thread 2: Counter = 2
Thread 2: Counter = 3
Thread 2: Counter = 4
Thread 2: Counter = 5
Thread 1: Counter = 6
Thread 1: Counter = 7
Thread 1: Counter = 8
Thread 1: Counter = 9
Thread 1: Counter = 10
Final Counter Value: 10
emertxe@emertxe-VirtualBox:~/Downloads/RTOS_Codes$ gcc binary_semaphore.c -o binary_semaphore -lpthread
emertxe@emertxe-VirtualBox:~/Downloads/RTOS_Codes$ ./binary_semaphore
Task1 Waiting for signal...
Task2 Sending Signal...
Task1 Received Signal!
emertxe@emertxe-VirtualBox:~/Downloads/RTOS_Codes$
```